BallTracking Module

1. Overview

The BallTracking module is responsible for detecting, tracking, and analyzing the trajectory of the ball in video sequences. It integrates with a fine-tuning YOLO called ball_yolo11x. detection with interpolation and shot-detection logic to deliver reliable ball coordinates across frames.

2.Dependencies

- **Ultralytics YOLO**: for real-time object detection.
- **OpenCV**: for geometric operations, polygon tests, and drawing.
- **Pandas**: for ball trajectory and interpolation methods.
- **Pickle**: for serializing detection results (stubs).

3.Class: BallTracking

Initialization

BallTracking(model_path: str, fps: int, confidence: float = 0.15)

- **Parameters**:
    1. model_path: Path to the YOLO model for ball detection.
    2. fps: Frames per second of the input video.
    3. Confidence: Minimum confidence threshold for YOLO detections. Default = 0.15

Methods

Interpolation Method

interpolate_ball(ball_positions, method='linear', order=3)

- **Purpose:** Fill missing ball positions in frames using interpolation methods.
- **Parameters:**
    1. ball_positions: List of dictionaries with ball detections per frame.
    2. method: Interpolation method ('linear', 'spline', 'polynomial', etc.).
    3. order: Polynomial, cubicspline or spline order (default 3).

- **Output:** List of dictionaries with interpolated ball positions per frame

Ball Hit Detection

get_shot_frames(ball_pos, changes=20)

- **Purpose:** Detect frames where the ball is hit based on changes in vertical movement.
- **Parameters:**
    1. ball_pos: List of dictionaries with ball detections per frame.

2. changes: Number of frame-to-frame directional changes to confirm a hit (default 20).

- **Output:** List of frame indices where ball hits are detected.

Detection Methods

detect_ball(frames, read_from_stub=False, stub_path=None*)*

- **Purpose:** Detect ball across all frames.
- **Parameters:**
  1. frames: List of video frames.
  2. read_from_stub: Load results from pickle if available.
  3. stub_path: Path to save/load detections.
- **Output:** List of ball detections per frame.

detect_frame(frame)

- **Purpose:** Run YOLO detection for a single frame.
- **Output:** Dictionary {1: bounding_box} for the detected ball.

Visualization Methods

draw_boxes(video_frames, src_detections, color_box=(255, 0, 0))

- **Purpose:** Draw bounding boxes and ball IDs on video frames.
- **Parameters:**
  1. video_frames: List of frames.
  2. src_detections: Detected ball positions per frame.
  3. color_box: Color of bounding box (default blue).
- **Output:** Annotated video frames.

4. Notes

- To use this module is important use the ball_yolo11x weights.
- This module integrates with **main.py**, where frames are first passed to YOLO player detection and tracking before ball detection.
- Works alongside PlayerTracking to provide synchronized data for rallies and shot analysis.
- Output ball coordinates and shot frames can be exported as CSV logs for post-match analysis.